

From Packets to Pixels: A Vision Transformer & Few-Shot Learning Approach to Malicious Network Threat Detection

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Abstract—Deep learning excels at deriving rich, high-dimensional features from modalities such as text, audio, and images. In network security, these methods offer a path to learn deep representations from the vast volumes of traffic that modern infrastructures carry. However, current malware-traffic analysis workflows rely on classical deep learning methods, such as convolutional or sequential architectures. While these methods are effective at capturing local byte-sequence patterns, they struggle to model dependencies that span long packet flows. Transformer models, with their use of self-attention, help address this limitation by attending globally to the input sequence. However, treating each network byte as an individual token leads to lengthy one-dimensional input sequences. In this paper, we introduce a compact, image-inspired encoding that converts contiguous byte segments into fixed patch sizes and arranges them on a two-dimensional grid. Each resulting patch functions as a single input token, dramatically reducing sequence length. Next, we leverage these packet images as input into a Vision Transformer (ViT) to efficiently attend across the entire packet. This patch-based representation preserves global context for self-attention while exploiting the ViT's strength in modeling spatially structured inputs. Lastly, to address the limitation of retraining the model to generalize to newly introduced malware classes, we introduce few-shot learning to quickly adapt to novel classes in a data-restricted environment. On the CIC-IoT23 malware-traffic dataset, our approach achieves impressive accuracy across multiple malware families, outperforming baseline methods.

Index Terms—Malware Classification, Traffic Classification, Vision Transformers, Few-Shot Learning

I. INTRODUCTION

As cyber threats continue to disrupt all forms of electronic communications, they have the potential to impact commerce [1], national security [2], and personal health and safety [3]. Exposure to these threats has escalated rapidly as the proliferation of connected devices and online data continues to expand. Cybercriminals have been quick to exploit developments in AI that empower them to create and distribute malicious attacks on an expanded scope with increased diversity and sophistication [4]. A powerful example of this ever increasing risk is evident in IoT devices that are consistently vulnerable to attacks due to their persistent availability on the network. The industrial IoT-cloud is a high-value target for attackers, making it susceptible to the deployment of malicious software, which can be exploited for adversarial purposes to undermine security [5].

One of the fundamental challenges in cybersecurity is the detection of network threats, and various methods have been reported in the field of network intrusion detection systems (NIDSs) [6]. However, as the access points are diversified, the

attack surface expands, thereby leaving the network systems vulnerable to potential threats. Moreover, cyberattack techniques have become more complex and sophisticated, and the frequency of attacks has also increased [6]. Attack frameworks are beginning to leverage the power of AI once they have achieved network penetration. This can be seen in the deployment of advanced metamorphic malware and ransomware [7]. By using AI-empowered obfuscation, an attacker can alter the internal structure of the malicious code with every attack. In the event that such malware gains access to the network, even if the original malware instance get detected, it can still infect the entire network. Current research and advances continue including the exploration of deep learning approaches, such as convolutional neural networks (CNNs), long short-term memory (LSTM), and autoencoders. Although these methods have demonstrated strong performance in detecting malicious threats, their deployment in real-world systems remains limited [6].

A promising direction for malware detection is transformer-based architectures [8], which use self-attention to capture dependencies in images without hand-crafted features [9]. Compared to CNNs, transformers provide a global receptive field from the first layer and scale well with data and model size, yielding strong performance on complex datasets and more interpretable attention patterns. However, since transformers were initially created for natural-language processing, they typically require lengthy one-dimensional sequences of packets to model global dependencies in packet data [10]. This often inflates training and inference time when every byte is treated as a token [10].

This paper explores an alternate method of network traffic classification for malware analysis that leverages the emerging power of Vision Transformers (ViT) [11] to capture contiguous bytes of network traffic into two-dimensional grids and train supervised models that can efficiently analyze the global context across entire packet flows and distinguish benign from malicious traffic. In our approach, we arrange packet bytes into two-dimensional “packet images” which allows the ViT to operate on a compact set of patch tokens rather than per-byte tokens, preserving global context provided by attention while avoiding the prohibitive sequence lengths that classical transformer-based architectures require for training. Given the speed and growing variability of cyberattacks, this proposed framework is then extended to extract representations of packet data from the pre-trained ViT and drive a Few-Shot Learning module [12] that reduces the need for annotated data and is

capable of generalizing to *novel malware classes* that were not present during ViT pre-training.

Our contributions are summarized below:

- **Novel Packet-to-Image Encoding.** We introduce a compact, image-inspired encoding, across several image pre-processing techniques, that converts contiguous byte segments into fixed-size patches arranged on a two-dimensional grid. This drastically reduces sequence length compared to packet byte-level tokenization, while preserving global packet context.
- **Vision Transformer for Malware Traffic.** We leverage a ViT backbone trained from scratch on these packet images to capture long-range dependencies with self-attention, outperforming classical deep learning baselines on multiclass malware-traffic benchmarks.
- **Few-Shot Learning Adaptation.** To handle emerging malware families with minimal data, we integrate a prototype-based few-shot module built on top of our frozen ViT features. Without full model retraining, this method achieves fast adaptation to novel classes, reducing annotation and data needs.

II. RELATED WORK

Deep Learning for Malware Detection has begun to be widely adopted for network security tasks, where these learned representations can often generalize quickly to known threats [13], [14]. Early approaches treated packet flow statistics as input and applied convolutional or recurrent models to capture local patterns and limited temporal context [15], [16]. While effective on short sequences, these architectures struggle to model long-range dependencies that arise across full packets or flows, and their focus on locality can miss global structure necessary for distinguishing closely related malware families. To address this, a line of work project raw packet bytes into two-dimensional layouts and employs 2D CNNs, leveraging spatial filters to exploit locality and regularities in the byte space [16]. Furthermore, more recent systems advocate end-to-end learning directly from payloads with minimal preprocessing, reporting competitive results across benchmark traffic datasets and operational settings [10], [17].

Transformers have begun to influence payload-based detection because self-attention can aggregate information over the entire input in a single layer. However, tokenizing each byte yields lengthy sequences and increased compute at inference time. In this work, we introduce ViTs as a malware feature extractor. This is a compelling alternative since we first arrange each packet into images. This reduces the token budget while preserving global context, enabling efficient, end-to-end learning over entire packets.

Few-Shot Learning seeks to adapt pre-trained models to novel classes from only a handful of labeled samples per class [12], [18]. In malware detection, this capability is essential because new families can emerge with limited labels, making full-model retraining time-consuming, resource-intensive, and

operationally impractical. Prototype-based methods, such as Prototypical Networks [12], form a per-class prototype as the mean of transformed support features, then assign unlabeled query samples to the closest class prototype under Euclidean distance. Recent work has explored few-shot methods for malware detection [19]–[21], but most rely on CNN-based feature extractors, which emphasize local patterns and struggle to capture long-range dependencies across entire sequences. To address this shortcoming, we implement the state-of-the-art ViT to globally attend to the packet. In similar work, researchers pre-train a transformer from scratch in a supervised manner on one-dimensional sequences and generalize to one novel class during their few-shot setup [22]. However, in our work, we perform a more robust experiment by pre-training the ViT on packet images from scratch on six malware classes, then use few-shot to generalize to three, never before seen malware classes.

III. DATA PROCESSING

A. Dataset Preparation

We processed network traffic data from the CIC-IoT23 dataset [23], which contains PCAP captures of both benign and malicious IoT network traffic across nine distinct attack categories. To address the computational requirements of Vision Transformer (ViT) models and the data scarcity challenges inherent in few-shot learning scenarios, we implemented a comprehensive preprocessing pipeline that transforms raw packet captures into structured image representations suitable for visual pattern recognition.

Each network packet's payload was extracted and truncated or zero-padded to a fixed length of 1500 bytes, corresponding to the standard Ethernet Maximum Transmission Unit (MTU). This standardization ensures consistent input dimensions across all samples while preserving the most informative portion of network traffic. To mitigate class imbalance, we employed balanced sampling with 12000 samples per class, randomly partitioned into training (70%), validation (15%), and test (15%) sets.

B. Image Encoding Methods

1) *RGB Hilbert Curve Encoding:* For the primary ViT experiments, we adopted a Hilbert curve-based encoding strategy that maps one-dimensional byte sequences onto two-dimensional 32×32 pixel grids while preserving spatial locality. The Hilbert curve, a space-filling fractal with order $\log_2(32) = 5$, ensures that adjacent bytes in the original payload remain proximate in the resulting image representation. Each byte value is normalized to $[0, 1]$ and encoded into RGB channels using differential scaling ($R = 1.0$, $G = 0.7$, $B = 0.5$), creating visually discriminative patterns that facilitate the ViT's self-attention mechanism in capturing both local and global traffic patterns.

2) *Grayscale Image Generation:* In addition to RGB Hilbert encoding, we investigated a simplified grayscale representation to evaluate the impact of channel complexity on

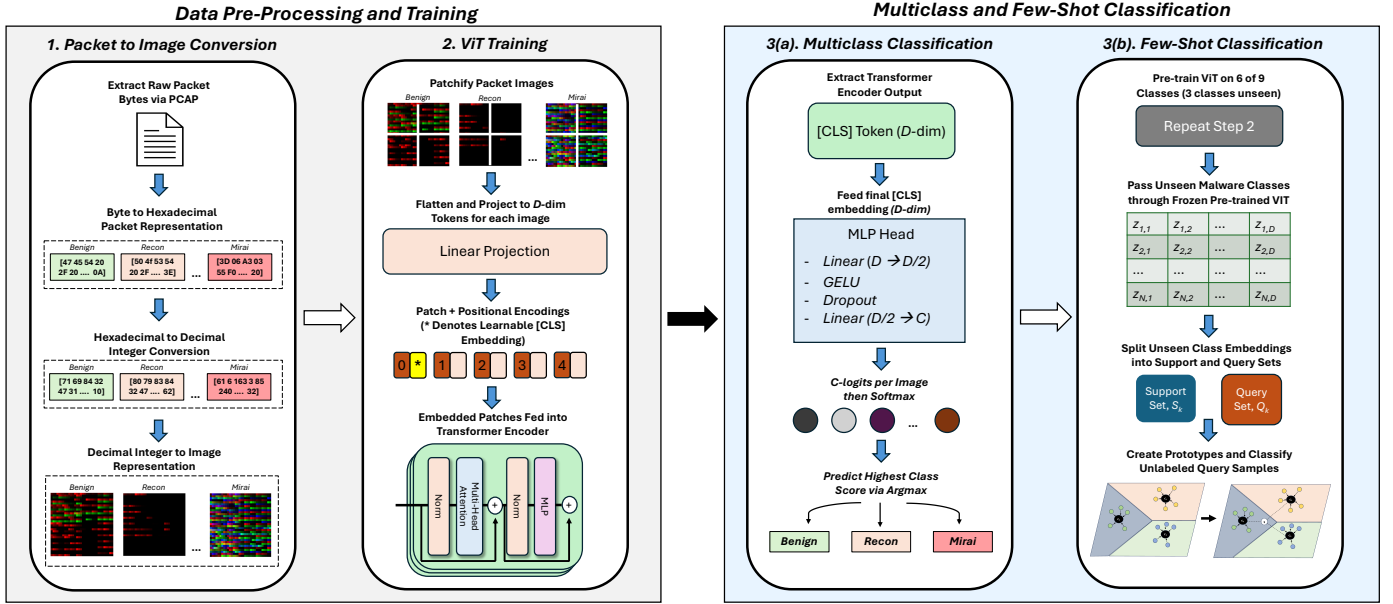


Fig. 1: The overall architecture of our proposed approach.

model performance. For grayscale images, the same Hilbert curve mapping was employed to transform the 1024-byte payload segments into 32×32 pixel grids. Each byte value (0–255) was directly normalized to the range $[0, 1]$ to create single-channel grayscale intensities. This approach reduces the input dimensionality from 3072 features (RGB) to 1024 features, potentially offering computational advantages while maintaining the spatial locality benefits of the Hilbert curve transformation. The grayscale encoding preserves byte-level patterns and relationships while eliminating the potential noise introduced by arbitrary channel assignments in RGB encoding.

These encoding approaches offer several advantages for transformer-based architectures, including: preservation of byte sequence relationships crucial for protocol structure recognition, efficient tokenization through the ViT’s patch embedding layer, and enhanced interpretability of learned attention maps. The resulting dataset enables efficient batch loading during both prototype network training and few-shot meta-learning episodes.

IV. METHODOLOGY

To address the limitations of existing approaches for malicious network traffic classification, we first propose a novel ViT-based framework for generalization in a multiclass malware detection scenario. Specifically, we apply the previously discussed data preprocessing techniques which result in a representative image for each packet traversing through a network. By transforming the packets into images, we can leverage the ViT’s strength in reducing the number of input tokens needed to train the model. Furthermore, this allows us to leverage the ViT’s ability to globally attend to the input image through positional-encodings and self-attention.

Building on this supervised pre-training, we further integrate a few-shot learning (FSL) module that allows rapid adaptation to previously unseen malware classes with only few examples. This two-stage design yields an intuitive and robust detector capable of improved accuracy on known classes and rapid extension to novel threats. The overall architecture of the proposed design is shown in Figure 1.

A. Vision Transformer

ViTs excel at capturing long-range dependencies with self-attention, making them well suited for our packet-image inputs. In this stage, we train a ViT backbone, denoted as $f_{\theta}(\cdot)$, **from scratch** on the packet images to extract generalizable features for multiclass malware detection. The model architecture follows the original ViT architecture, allowing for a scalable and efficient implementation.

Patch Embedding. Let each packet image be denoted as $I \in \mathbb{R}^{H \times W \times C_{in}}$, where C_{in} is the number of input channels and H, W are the spatial dimensions. I is first split into $N_p = \frac{HW}{P^2}$ non-overlapping patches of size $P \times P$. These patches are each flattened into a vector of length $P^2 C_{in}$. A single trainable linear layer then projects each $P^2 C_{in}$ -dimensional vector into a D -dimensional embedding using $\mathbf{W}_E \in \mathbb{R}^{(P^2 C_{in}) \times D}$, yielding N_p vectors of size D that are fed into the transformer.

A learnable [CLS] embedding $\mathbf{x}_{cls} \in \mathbb{R}^D$ is prepended to the sequence of patch embeddings, increasing the effective sequence length to $N_p + 1$. To retain spatial information, learnable one-dimensional positional embeddings $\mathbf{E}_{pos} \in \mathbb{R}^{(N_p + 1) \times D}$ are added. The resulting sequence is:

$$\mathbf{z}_0 = [\mathbf{x}_{cls}; \mathbf{x}_p^1 \mathbf{W}_E; \mathbf{x}_p^2 \mathbf{W}_E; \dots; \mathbf{x}_p^{N_p} \mathbf{W}_E] + \mathbf{E}_{pos}. \quad (1)$$

Transformer Encoder. The Transformer encoder is composed of a stack of L identical layers. Each layer consists of two main sub-layers: a multi-head self-attention (MSA) block and a position-wise, fully connected feed-forward network (MLP), as illustrated in Figure 1.

The self-attention mechanism, a key component of the Transformer, allows the model to weigh the importance of different patches in the input sequence. For each patch embedding in an input sequence $\mathbf{z} \in \mathbb{R}^{(N_p+1) \times D}$, we compute query ($\mathbf{q}$), key ($\mathbf{k}$), and value ($\mathbf{v}$) vectors by projecting the sequence with learned weight matrices $\mathbf{W}_{qkv}$. The attention weights are then calculated based on the scaled dot-product similarity between queries and keys. The standard qkv self-attention (SA) is defined as:

$$[\mathbf{q}, \mathbf{k}, \mathbf{v}] = \mathbf{z} \mathbf{W}_{qkv}, \quad \mathbf{W}_{qkv} \in \mathbb{R}^{D \times 3d_h}, \quad (2)$$

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{q}\mathbf{k}^T}{\sqrt{d_h}}\right), \quad \mathbf{A} \in \mathbb{R}^{(N_p+1) \times (N_p+1)} \quad (3)$$

$$\text{SA}(\mathbf{z}) = \mathbf{A}\mathbf{v}, \quad (4)$$

where d_h denotes the per-head dimensionality. Our model employs multi-head self-attention, which runs multiple self-attention operations, or “heads,” in parallel and projects their concatenated outputs. This allows the model to jointly attend to information from different representation subspaces. Our specific configuration, including the number of attention heads, is detailed in Table I.

A LayerNorm (LN) operation is applied before each block, and residual connections are applied after each block. The self-attention layers are global, which gives the ViT a significant advantage over traditional CNNs by allowing it to integrate information across the entire image from the very first layer. The operation of each encoder layer for $\ell = 1 \dots L$ is defined as:

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad (5)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell. \quad (6)$$

Classification Head. As shown in Fig. 1 (3a), the final classification is derived from the state of the [CLS] token at the output of the Transformer encoder, $\mathbf{z}_L^0 \in \mathbb{R}^D$. This D -dimensional vector serves as the aggregate representation of the entire packet image and is passed to an MLP head: a linear layer $D \rightarrow D/2$ with GELU and dropout, followed by a final linear layer mapping to C logits (number of traffic classes). A softmax is applied to the logits to produce the final class probabilities.

Model Configuration and Training. The Vision Transformer was built and trained using the specific hyperparameters detailed in Table I. The model was trained for 30 epochs with a batch size of 32. We used the AdamW optimizer [24] with a learning rate of 1e-4 and a learning rate scheduler. The loss was computed using a weighted Cross-Entropy Loss function.

TABLE I: ViT Model Configuration

Hyperparameter	Value
Image Size	32×32
Channels (C_{in})	3
Patch Size (P)	16×16
Embedding Dim (D)	192
Transformer Layers (L)	6
Attention Heads	3
Number of Classes (C)	9

B. Few-Shot Learning Adaptation

In our FSL scenario, we first pre-train the ViT, denoted as $f_\theta(\cdot)$, on a subset of six malware classes, deliberately excluding three classes to be used for few-shot evaluation. Holding out these classes during pre-training is critical for our study as this prevents information leakage and forces the model’s downstream few-shot adaptation to generalize to unseen malware families. This setup represents a realistic threat setting where emerging malware classes appear after model deployment and the defender must rapidly adapt using a few labeled examples of the novel introduced malware class.

Episodic Task Construction and Prototype Extraction.

After pre-training the ViT in a supervised manner on a subset of six malware classes, we perform episodic few-shot evaluation on the three held-out classes using an episodic few-shot evaluation protocol. Each episode is randomly sampled and constructed as an N -way, K -shot classification task: we sample N classes, and for each class, we draw K labeled support examples. These form the support set $\mathcal{S} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{NK}$, where NK is the total number of labeled support samples, $\mathbf{x}_i \in \mathbb{R}^{H \times W \times C_{\text{in}}}$ denotes a *packet image from the novel class set*, and $y_i \in \{1, \dots, N\}$ is the corresponding class label. Additionally, we sample U query examples per class to form the query set $\mathcal{Q} = \{(\mathbf{x}_q, y_q)\}_{q=1}^{NU}$. The query labels are withheld from the model and used only to evaluate performance.

Support examples are embedded by $f_\theta(\cdot)$ to form class prototypes, $\mathbf{c}_k$, by taking the mean of the transformed feature vectors of the support set embedding [12]. We first compute support embeddings $\mathbf{z}_i = f_\theta(\mathbf{x}_i) \in \mathbb{R}^D$ and formalize the class prototypes as:

$$\mathbf{c}_k = \frac{1}{K} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_k} \mathbf{z}_i, \quad (7)$$

where $\mathcal{S}_k \subset \mathcal{S}$ is the subset of support examples belonging to class k . Similarly, for each query $\mathbf{x}_q$, we compute its embedding $\mathbf{z}_q = f_\theta(\mathbf{x}_q)$.

Prototype-based Few-Shot Classification. Given the class prototypes $\{\mathbf{c}_k\}_{k=1}^N$ and a query embedding $\mathbf{z}_q = f_\theta(\mathbf{x}_q)$, we compute the squared Euclidean distance between each query and prototype as $d(\mathbf{z}_q, \mathbf{c}_k) = \|\mathbf{z}_q - \mathbf{c}_k\|^2$ and treat $-d(\mathbf{z}_q, \mathbf{c}_k)$ as the logit for class k . These logits are converted into a distribution over classes by applying a softmax function:

$$p(y = k | \mathbf{x}_q) = \frac{\exp(-d(\mathbf{z}_q, \mathbf{c}_k))}{\sum_{k'} \exp(-d(\mathbf{z}_q, \mathbf{c}_{k'}))}. \quad (8)$$

TABLE II: Results of Multiclass Classification of Malware Types on the CIC-IoT23 Test Dataset

Method	Accuracy	Precision	Recall	F1-Score
1D-CNN [15]	80.48	84.03	80.48	81.42
2D-CNN [16]	93.73	93.86	93.73	93.78
LSTM [16]	90.59	91.27	90.59	90.76
ViT + Grayscale	91.24	92.10	91.20	91.50
ViT + RGB Hilbert	93.81	94.20	93.80	93.90

so that prototypes closer to the query embedding receive a higher probability. The predicted label for $\mathbf{x}_q$ is $\arg \max_k p(y = k | \mathbf{x}_q)$. Since $f_\theta(\cdot)$ remains frozen (no gradients are back-propagated) during our few-shot testing scenario, we simply compute accuracy over the query samples for each episode and report the average over all tested episodes.

V. EXPERIMENTS AND RESULTS

We evaluated our approach on the CIC-IoT23 dataset with 12,000 samples per class, split 70/15/15 for training, validation, and testing. For few-shot evaluation, we pre-train the ViT on 6 selected malware classes and generalize to 3 newly introduced malware classes under 1-,2-,5-, and 10-shot scenarios.

A. Multiclass Classification Results

Table II presents the performance of our ViT-based approach against traditional baselines on multiclass malware classification. Our ViT with RGB Hilbert encoding achieves **93.81% accuracy**, outperforming most baseline methods and demonstrating the effectiveness of transformer architectures for malicious network traffic classification.

The 1D-CNN struggles to capture spatial relationships in packet bytes, achieving only 80.48% accuracy due to the inability to model global dependencies across the entire packet sequence. The 2D-CNN performs better at 93.73% by leveraging spatial convolutions on packet representations, but remains limited by local receptive fields that cannot capture long-range packet dependencies. The LSTM achieves 90.59% accuracy but suffers from sequential processing bottlenecks and gradient vanishing issues with long packet sequences. Our approach overcomes these limitations through global self-attention, enabling simultaneous attention to all regions and superior feature learning from the packet images.

To further validate this discriminative capability, Figure 2 provides detailed per-class breakdown of our ViT model's classification performance, demonstrating the model's ability to learn discriminative features from Hilbert-encoded packet representations. The strong diagonal performance validates our core hypothesis that vision transformers can effectively learn from packet-to-image representations. The model excels at flooding-based attacks, with DDoS-HTTP Flood (97.9%) and DoS-UDP Flood (97.8%) achieving near-perfect accuracy due to their distinctive volumetric signatures in Hilbert-encoded images.

The confusion matrix reveals a triangular confusion pattern among Recon-PortScan (83.8%), DictionaryBruteForce

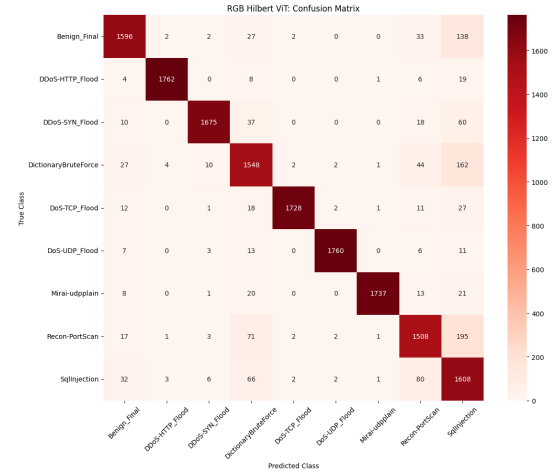


Fig. 2: Confusion matrix for ViT with RGB Hilbert encoding.

TABLE III: Few-shot Learning Accuracy (%) across Different Shots

Model	1-shot	2-shot	5-shot	10-shot
1D-CNN (Raw Bytes)	55.44	56.49	57.86	59.56
2D-CNN (RGB Hilbert)	81.67	90.33	94.35	94.64
LSTM (RGB Hilbert)	77.06	82.64	87.39	88.39
ViT (Grayscale)	81.35	87.99	92.28	92.95
ViT (RGB Hilbert)	80.79	90.78	94.35	94.81

(86.0%), and SqlInjection (89.3%), with 195 and 162 cross-misclassifications respectively. These attacks rely on subtle payload manipulations rather than volumetric patterns, making them harder to distinguish in visual representations. The model achieves strong benign traffic classification (88.7%), critical for minimizing false alarms, and these learned representations serve as the foundation for our few-shot learning approach.

B. Few-Shot Learning Results

Table III highlights our method's capacity to quickly generalize to previously unseen malware families using only a few labeled examples. For fairness, all feature extractors were pre-trained on the same six malware classes as the ViT, with the same three classes withheld for evaluation over 1000 few-shot episodes. Each architecture then serves as the feature extractor for its respective experiment. Our ViT with RGB Hilbert encoding achieves **94.81% accuracy** in 10-shot scenarios, showcasing effective knowledge transfer capabilities.

The 1D-CNN's poor few-shot performance (59.56% at 10-shot) stems from its reliance on local sequential patterns that fail to generalize across malware families with limited training data. The 2D-CNN achieves competitive results (94.64% at 10-shot), nearly matching our ViT approach at 94.81%. This minimal difference suggests both architectures extract similarly effective features for prototype-based classification, though ViT demonstrates superior supervised learning performance (96.94% vs 93.73%) when sufficient training data is available. The LSTM plateaus at 88.39% due to sequential processing

limitations that hinder effective prototype formation. The RGB Hilbert encoding preserves critical spatial relationships that benefit both CNN and ViT architectures in few-shot scenarios.

C. Analysis and Discussion

Our results demonstrate two key advantages of the proposed ViT based approach. First, the RGB Hilbert encoding consistently outperforms grayscale variants across both experimental settings, indicating that multi-channel representations better preserve packet byte relationships critical for malware discrimination. Second, our method achieves rapid adaptation to novel threats improving from 80.79% (1 shot) to 94.81% (10 shot) a critical capability in cybersecurity where new malware variants emerge continuously. The combination of global attention modeling and compact image representations enables both strong supervised performance and effective few shot generalization without requiring full model retraining for emerging threat detection.

VI. CONCLUSION AND FUTURE WORK

In this paper, we introduced a novel two-stage framework for malicious network threat detection that addresses the limitations of traditional deep learning models. By converting raw network packets into compact image representations using an RGB Hilbert curve encoding, we enabled the use of a Vision Transformer (ViT) to effectively capture long-range dependencies across entire packet flows. Our results demonstrate that this approach outperforms conventional 1D-CNN, 2D-CNN, and LSTM architectures in a multiclass malware classification task, achieving a peak accuracy of **93.81%**. Furthermore, to tackle the critical challenge of detecting new and emerging threats, we integrated a prototype-based few-shot learning module. By leveraging the rich feature embeddings from the pre-trained ViT, our model can rapidly adapt to novel, previously unseen malware classes with minimal labeled examples. This was validated by our few-shot experiments, where the model achieved **94.81%** accuracy with only ten samples per new class. Our work successfully demonstrates a robust, scalable, and adaptable solution that enhances malware detection capabilities in the face of an evolving cyber threat landscape. Building on the promising results of this study, several avenues for future research can be explored, such as implementing more advanced image encoding techniques, validate the model's effectiveness in high-stakes Industrial Internet of Things (IIoT) and critical infrastructure environments sectors, such as power grids or transportation systems. This will help assess its suitability for protecting operational technology (OT) networks.

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